

Processing of grammatical information in cochlear implant simulated speech by German adult listeners

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Abstract

This study investigates how German which-questions are processed in cochlear implant (CI) simulated speech compared to normal speech. In addition, it explores the role of working memory capacity (WM). Using eye-tracking, accuracy and gaze patterns for subject, object and passive questions were studied. Questions were disambiguated with grammatical information, specifically morphosyntactic cues: case and/or subject-verb agreement. Results show that only object question accuracy was affected by CI simulations. Similarly, eye movements showed weaker interpretation preferences for object questions in CI simulations. Participants with high WM had higher accuracy and their eye gaze showed quicker revisions of garden path effects compared to those with medium or low WM capacity. These findings show that morphosyntactic cues support correct interpretation of object questions in CI simulations, but processing these questions is slower and influenced by working memory capacity.

Index terms: Sentence comprehension, cochlear implant simulated speech, working memory

1. Introduction

1.1. Morphosyntactic cues

Language contains morphosyntactic cues, like case and subject-verb agreement, that indicate grammatical dependencies within sentences and are thus relevant for interpretation. Some of these cues are not very salient and might easily be missed, especially in adverse listening conditions, like hearing with a cochlear implant. Schouwenaars et al. [1] found that German children with cochlear implants relied less on these cues in sentence processing, possibly due to reduced speech exposure or the quality of cochlear implant speech. The goal of the current study is to investigate this last explanation, i.e., CI signal quality. We use cochlear implant simulated speech to understand an effect on language processing in children with a CI. Furthermore, the role of working memory capacity is investigated as it affects sentence processing, with deviations observed in cochlear implant users.

We examined the interpretation of wh-questions, that enable us to examine different morphosyntactic cues, namely case and subject-verb agreement. In German, the constituent structure of object questions (NP-V-NP) look identical to those of subject questions (NP-V-NP), making them structurally ambiguous and difficult to understand for the listener. Nevertheless, morphosyntactic cues can serve to distinguish between subject and object questions, aiding in accurate interpretation. Nominative and accusative case marking identify subjects and objects, illustrated in the subject question (1) and object

questions (2) in Table 1. The forms *welcher* and *der* of singular masculine nouns in German indicate subjects, whereas *welchen* and *den* indicate objects. Note that feminine and plural nouns do not distinguish subjects and objects, instead subject-verb agreement may also disambiguate as in (3). In subject question (3), agreement in number between the first NP and the verb indicates a subject question. In object questions (4), agreement between the second NP and the verb, rather than the first, indicates an object question. A combination of cues, like in (5) and (6), is also possible.

Table 1: Examples of subject and object wh-questions disambiguated by case (*nom* = nominative; *acc* = accusative; *amb* = ambiguous case) or subject-verb agreement (*sg* = singular; *pl* = plural).

Subject question: Case					
1)	Welcher	Panda	wäscht	den	Tiger?
	Which	Panda	wash	the	Tiger?
	<i>nom</i>	<i>sg</i>	<i>sg</i>	<i>acc</i>	<i>sg</i>
'Which Panda is washing the Tiger?'					
Object question: Case					
2)	Welchen	Panda	wäscht	der	Tiger?
	Which	Panda	wash	the	Tiger?
	<i>acc</i>	<i>sg</i>	<i>sg</i>	<i>nom</i>	<i>sg</i>
'Which Panda is the Tiger washing?'					
Subject question: Agreement					
3)	Welche	Katze	wäscht	die	Schildkröten?
	Which	cat	wash	the	turtles?
	<i>amb</i>	<i>sg</i>	<i>sg</i>	<i>amb</i>	<i>pl</i>
Object question: Agreement					
4)	Welche	Katze	waschen	die	Schildkröten?
	Which	cat	wash	the	turtles?
	<i>amb</i>	<i>sg</i>	<i>pl</i>	<i>amb</i>	<i>pl</i>
Subject question: Agreement and Case					
5)	Welche	Frösche	waschen	den	Vogel?
	Which	frogs	wash	the	bird?
	<i>amb</i>	<i>pl</i>	<i>pl</i>	<i>acc</i>	<i>sg</i>
Object question: Agreement and Case					
6)	Welche	Katze	wäscht	der	Vogel?
	Which	Panda	wash	the	bird?
	<i>amb</i>	<i>pl</i>	<i>sg</i>	<i>nom</i>	<i>sg</i>

Even though word order cannot be used to distinguish between subject and object questions in German and other languages like Dutch and Italian, it still plays a role in sentence interpretation. More specifically, we tend to interpret the first part of the sentence as the subject, or performer of the action. When the assumed subject does not agree with the verb, listeners are misled (garden path) and must revise their interpretation. This occurs in object questions, explaining subject-object asymmetries in acquisition and processing. More specifically, subject-first sentences are acquired earlier and processed quicker compared to object-first sentences.

Revising garden paths seems linked to WM [2], and thus the ability to temporarily store and process information. It has been argued that people with larger WM keep more interpretations active, whereas people with smaller capacities only the more likely interpretation [3]. Others argue that people with smaller WM have problems to suppress dispreferred representations as effectively as people with larger WM [4].

1.2. Cochlear implant simulated speech

In normal hearing, thousands of hair cells capture a wide range of sounds and transmit signals to the auditory nerve. In contrast, cochlear implant users rely on a maximum of 22 electrodes to stimulate the auditory nerve [5], significantly limiting sound resolution [6]. Additionally, cochlear implant users experience a significantly reduced dynamic range – the difference between barely audible and maximum comfortable sound levels – typically around 10-20 dB compared to approximately 100 dB in normal hearing [5]. As a result, their language input is less rich than that of normal-hearing listeners. To investigate the effects of this degraded speech input, this study uses noise-vocoded speech to simulate cochlear implant processing in normal-hearing participants.

1.3. Current study

We predict that participants with CI simulations will perform worse on object questions than those with normal speech, while performance on subject and passive questions will remain unaffected. As described above, correct interpretation of object questions requires the use of morphosyntactic cues, which may be hard to perceive in sentences in CI simulated speech. In contrast, subject questions can be accurately interpreted just on a subject-first preference (see Table 1). Likewise, passive questions depend on more and perceptually salient cues, such as passive verb morphology for correct interpretation (e.g., *Welcher Tiger wird von dem Panda gewaschen?* Which tiger is being washed by the panda?). Additionally, we predict that working memory capacity influences the processing of object questions.

2. Method

2.1. Participants

Two groups of thirty German-speaking young adults between 19 and 30 years old without hearing loss participated in this study. The test group (19 female, mean age: 24) received the stimuli in cochlear implant simulated speech (CISS) and the control group (16 female, mean age: 24) received the stimuli in normal speech (NS). Before starting the experiment, hearing was examined with a pure tone audiogram. Participants gave written consent prior to participating and the study was approved by the Ethical Committee of University of Oldenburg and in accordance with the declaration of Helsinki.

Four participants, all from the test group, did not pass the screening tests and were excluded for further analysis. Therefore, the final test group consisted of 26 participants (16 female, mean age: 25).

2.2. Design and materials

Two screening tests, a digit span task and a which-questions comprehension test were administered.

2.2.1. Screening tests

The screening tests consisted of an auditory discrimination task testing case marking differences and a picture-selection task testing the comprehension of subject-verb agreement in canonical declarative sentences. Both screening tests consisted of 16 items and participants passed this test when they scored correctly on 14 or more items. As can be seen in examples (1-4), morphosyntactic cues are not very salient. For this reason, screening tests were conducted to be certain that the participants could distinguish between nominative and accusative case and that they did use verbal inflection to identify the number of the subject in canonical declarative sentences.

2.2.2. Digit span task

A digit span task was administered to test participants working memory. The experimenter would read out loud a sequence of digits ranging from 1 to 9. In the first condition, forward, participants recalled this sequence in the same order. In the second condition, backward, they recalled it in the reversed order. The number of digits in the longest sequence recalled correctly in the reversed order, the backward span, was used as a measure of working memory as in addition to temporary storage, it also involves manipulation of information [7].

2.2.3. Which-questions comprehension test

The which-questions comprehension test was a picture selection task with eye-tracking, including subject, object and passive questions. Subject and object questions were disambiguated either by only case on both NPs (*Case*), by only subject-verb agreement (*Agr*) or by both subject-verb agreement and case on the second NP (*AgrCa*). These conditions were created considering gender and number of nouns. In particular, German singular masculine nouns (but not feminine nouns) distinguish between nominative (*der*) and accusative (*den*) case. Alternating the number between the two NPs allowed subject-verb agreement to serve as a cue. Regarding the timing of the disambiguation, the *Case* condition provides an early cue at the *wh*-word, whereas in *Agr* and *AgrCa* conditions, disambiguation occurs later at the verb. Passive questions contained the same nouns but were always disambiguated by passive morphology (the verb *werden*, the past participle, and the *by*-agent). Passives were included as a comparative condition. As in object questions, the patient proceeds the agent. But, unlike in active object questions, in passive questions the agent is expressed as the subject.

Four lists varied in item order and target position, each containing 54 test items: 18 per question type and six per cue condition. Six transitive verbs were used, with distinct singular and plural forms (e.g., *fängt-fangen* ‘catches-catch’). Fifteen noun pairs were selected based on a questionnaire study to ensure plausibility and to control for the influence of semantic properties on thematic role assignment (i.e., a donkey washing a panda was evaluated as plausible as the reverse). Sentences were prerecorded by a female native speaker of German.

In each trial, two pictures, depicting two reversed interpretations were displayed next to each other. To illustrate, in Figure 1, the left picture represents the correct object-first interpretation of sentence (2) and the right picture the incorrect subject-first interpretation.

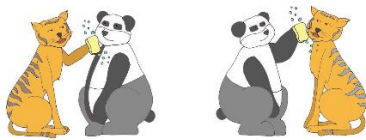


Figure 1: An example of a picture pair, with one illustrating the object-first interpretation (left) and the other depicting the subject-first interpretation (right) of sentence (2): Welchen Panda wäscht der Tiger? 'Which panda is the tiger washing?'.

The test items were divided into two blocks of 27 test items, each preceded by two practice items and including seven fillers. Participants were seated in front of a Tobii TX300 eye tracker, connected to two computers. One ran the experiment using E-Prime 2.0 (Psychological Software Tools, Inc.) and collected behavioral data, while the second recorded gaze data at a sample rate of 300 Hz using TET-calls in E-Prime.

2.2.4. Cochlear implant simulated speech

For the simulation, the sentences were processed through a software-implemented vocoder (adapted from [8]; for technical details see [9,10]). This vocoder simulates the signal processing mechanisms and the physiology of cochlear implants users. It was configured to simulate an implant of the brand Cochlear Ltd. with a Contour Advance electrode array featuring 22 electrodes and included both a CI simulation and an auralization component, and accounted for channel interaction.

2.3. Procedure

Firstly, a pure tone audiogram was administered and the digit span test was performed. After that, the participants of the test group had a small familiarization session of the CI simulations, in which they heard a two-minute fairytale twice and additionally 8 sentences comparable to the actual test sentences. Then, the first half of the which-questions test was conducted, followed by the agreement-screening test. After which, the second half of which-questions test was conducted, followed by the case screening test. All speech content was transmitted through Sennheiser HD650 headphones. Altogether, the procedure lasted approximately one hour.

2.4. Statistical analysis

Generalized linear mixed-effects regression models (GLMER) with the software R (version 3.1.2) were used to analyse the accuracy data. For variables with more than two values, multiple comparisons were made using the 'multcomp' package [11].

The first analyses compared the two groups (CISS vs. NS) to examine the effect of CI simulations on which-question comprehension. Accuracy was the dependent variable, with *Participant* and *Item* as random intercepts. Random slopes were evaluated but did not improve the model. Sequentially, fixed factors (*Group*, *Question Type*, *Cue Type*) were added to assess model fit using Akaike-Information-Criterion [12]. *Cue Type* did not significantly improve the model, nor were any interactions found with *Group* or *Question Type*. Additionally, potential effects of material-related variables (*Verb*, *Noun Pair*, *Version*, *Direction of Action* and *Position of Target*) were examined but did not significantly improve the model.

The second GLMER model focused on object questions in CI simulations, investigating potential effects like WM. Accuracy on object questions was the dependent variable, with *Item* the only random factor, as there was one digit span score per participant. *Trial number* and *digit span* were included as fixed factors.

Generalized Additive Mixed Modeling (GAMM) [13], was used to analyze gaze data with *mgcv* 1.8.4 [13] and *itsadug* [14] packages. *TCDiff* (difference between looks towards the target and competitor) for 200 ms timebins was the dependent variable. Interactions between *Group* (SCIS vs. NS) and *Type of Question* (subject vs. object vs. passive) were combined into a single predictor to enable assessing group differences across question types. *Participant* and *Type of Question* were merged into a random effect factor (*ParticipantQuestion*). Difference plots and the *get_differences* function were applied to calculate differences between conditions.

3. Results

3.1. Overall accuracy

Figure 2 displays the percentage of correct interpretations of the different type of which questions in CISS and NS.

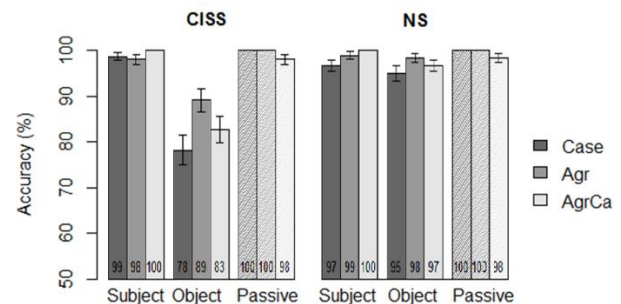


Figure 2: Percentages of correct interpretations of subject questions, object questions and passive questions.

In the final model an interaction between *Group* and *Type of Question* revealed that participants with CI simulated speech performed worse than those with normal speech on object questions ($\beta = -2.08$, $z = -4.34$, $p < .001$), but not on subject and passive questions. Furthermore, accuracy on object questions was significantly lower than on subject ($\beta = -3.62$, $z = -5.55$, $p < .001$) and passive questions ($\beta = 4.15$, $z = -5.48$, $p < .001$) for CISS participants, with no difference between subject and passive questions. Finally, no significant differences in accuracy were found for NS.

3.2. Accuracy on object questions and working memory

For the analysis of accuracy scores on object questions by participants presented with CISS, both trial number and backward digit span were significant predictors, and an interaction between these two variables was detected. There were more incorrect responses in the early trials compared to later ones ($\beta = 0.46$, $z = -3.50$, $p < .001$). There was also an effect of digit span: higher digit span scores were associated with higher accuracy scores on object questions ($\beta = 0.68$, $z = 3.15$, $p < .01$). Because trial number influenced accuracy, the digit span only affected earlier trials, not the later ones, explaining the interaction.

3.3. Eye movements

Whereas accuracy reveals final interpretations, eye-tracking provides insight into interpretation preferences during sentence processing. Figure 3 shows gaze patterns for both groups for subject, object and passive questions.

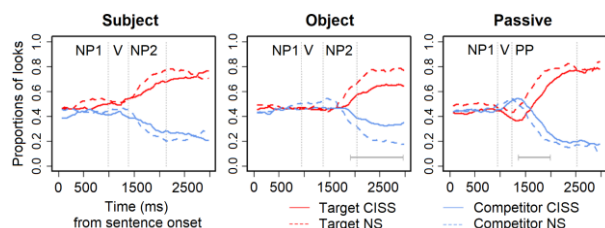


Figure 3: Online gaze behavior of the CISS (solid) and NS (dashed) groups for subject, object and passive questions. Red and blue lines show gazes to the target and competitor picture, respectively. Horizontal grey lines indicate significant group difference (CISS vs. NS) based on the statistical model.

For subject questions, both groups increasingly look towards the target picture, with no significant differences between them. In object questions, the NS group initially looks more at the competitor picture, whereas the CISS group does not, illustrating weaker preferences for this group. For passive questions, both groups first look at the incorrect interpretation, but the CISS group take longer to revise their interpretation.

Summarizing, CI simulated speech appears to affect the online processing of object questions by resulting in less clear interpretation preferences during the processing of object questions. It also leads to longer revision periods for incorrect initial interpretations in passive questions.

Working memory affected object question accuracy. Here, we examine its impact on online processing, focusing on object questions. Whereas no clear interpretation preferences emerged for the CISS group when looking at the object questions as a whole, a closer examination reveals initial incorrect interpretation preferences in cases where disambiguation occurred later in the sentence at the verb, i.e. the conditions where verb agreement provides a cue (*Agr* and *AgrCa*). Questions disambiguated by case were excluded, as they disambiguate right from the start.

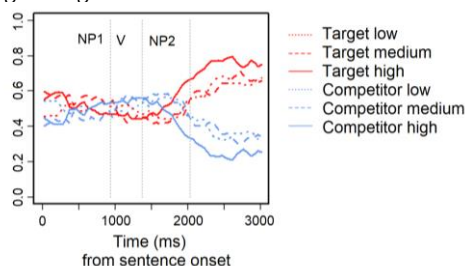


Figure 4: Online gaze behavior for object questions in the *Agr* and *AgrCa* conditions in CISS, showing looks towards the target (red) and competitor (blue) pictures across Digit Span Groups: low (dotted), medium (dashed), and high (solid).

Figure 4 shows the gaze patterns of temporarily ambiguous object questions (*Agr* and *AgrCa*) across digit span groups. Regardless of whether participants had low (span: 3-4), medium (span:5) or high (span: 6-7) span scores, all show an initial

preference for the subject-first interpretation. Notably, participants in the high span group display this preference earlier, switch interpretations sooner, and have more looks to the target picture at the end compared to participants in the medium or low span groups.

4. Discussion

The current study found that CI simulated speech affects the interpretation of object questions, but not subject or passive questions, as predicted. In subject questions, using a subject-first preference leads to the correct interpretation. For passive questions, passive morphology guides interpretation. Object questions require morphosyntactic cues, such as case marking and subject-verb agreement, which are challenging to perceive in CI simulated speech, as discussed below.

One object question item, *Welchen Maler fängt der Cowboy?*, was particularly difficult in CI simulated speech, but not in normal speech. The accusative case marking, ending in a nasal *n* followed by a noun starting with a nasal *m*, may have been unclear, leading some participants to mistake it for nominative case due to phonological assimilation. The remaining case cue, *der* in the second NP, might not have been perceived strongly enough or considered a strong enough cue for a revision of the subject-first interpretation. This highlights how spectral degradation in CI simulated speech can hinder interpretation.

Trial number significantly predicted object question accuracy, with more object questions misinterpreted as subject questions in earlier trials, compared to later trials. This again highlights the vulnerability of morphosyntactic cues in CI simulated speech. Subject and passive questions were understood correctly from the start of the experiment, suggesting that noun phrases and voice were more easily perceived. The higher error rate in object questions at the beginning of the experiment were therefore not solely due to participants adjusting to CI simulated speech. Instead, it seems as if participants needed time to adjust to object-first structures and rely more on morphosyntactic cues than on word order preferences.

Overall, participants in the CISS group with higher working memory capacities performed better on early object-question trials compared to those with lower working memory capacities. Regarding the debate on WM and the number of active representations during processing [4], our gaze data shows that all WM groups favor one most likely interpretation. Regardless of low, medium or high working memory capacities, participants' gaze patterns indicate a preference for the subject-first interpretation in temporarily ambiguous object questions. Additionally, adults with high WM revise the subject-first interpretation earlier and more confidently compared to adults with medium or low WM, as reflected in their quicker and more frequent gazes toward the target.

Similar to temporarily ambiguous object questions, passive questions are initially interpreted as subject questions. In CI simulated speech, revising this incorrect interpretation takes longer compared to normal speech. This emphasizes the additional processing time required to revise incorrect interpretations in degraded speech processing.

To conclude, the processing of morphosyntactic cues in wh-questions is affected by CI simulated speech. Therefore, degraded speech input might partly explain comprehension difficulties in children with CIs. Future research could explore CI simulated speech in normal-hearing language-learning children.

5. References

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